

Data Structures & Algorithms

Quiz: Singly Linked List

Answer all questions. For code-based questions, assume the standard node declaration unless stated otherwise.

1. Each node of a singly linked list contains:
 - (a) Data and a pointer to the previous node
 - (b) Data and a pointer to the next node
 - (c) Data only
 - (d) Pointers to both the previous and the next node
2. The next field of the last node in a singly linked list stores:
 - (a) The address of the head node
 - (b) The address of itself
 - (c) NULL
 - (d) A garbage value
3. A structure that contains a pointer to a variable of its own type is called:
 - (a) Nested structure
 - (b) Self-referential structure
 - (c) Recursive union
 - (d) Anonymous structure
4. A singly linked list is empty when:
 - (a) `head->next == NULL`
 - (b) `head == NULL`
 - (c) `head->data == 0`
 - (d) `head->next == head`
5. Which of the following correctly inserts newnode at the beginning of a list with head pointer head?
 - (a) `head = newnode; newnode->next = head;`
 - (b) `newnode->next = head; head = newnode;`
 - (c) `newnode = head; head->next = newnode;`
 - (d) `newnode->next = head->next; head = newnode;`
6. Which statement sequence correctly deletes the first node of a non-empty list?
 - (a) `free(head); head = head->next;`
 - (b) `temp = head; head = head->next; free(temp);`
 - (c) `head = head->next; free(head);`
 - (d) `temp = head->next; free(head); head = temp->next;`
7. Which operation is **not** directly possible in a singly linked list?
 - (a) Traversing from head to tail
 - (b) Inserting a node after a given node
 - (c) Traversing from tail to head
 - (d) Deleting the head node
8. In a singly linked list of n nodes, the total number of next pointer fields is:
 - (a) $n - 1$
 - (b) n
 - (c) $n + 1$
 - (d) $2n$

9. Given a pointer p to a node that is **not** the last node, and no access to the head pointer, the node can be deleted by:
- (a) Calling free(p) directly
 - (b) Setting p = NULL
 - (c) Copying the data of p->next into p, then deleting p->next
 - (d) It is impossible under any circumstances
10. The correct loop condition to traverse and print all nodes using a pointer p initialised to head is:
- (a) while (p->next != NULL)
 - (b) while (p != NULL)
 - (c) while (p->data != NULL)
 - (d) while (p != head)
11. To insert a node at the end of a singly linked list, traversal must continue until reaching the node where:
- (a) p->data == NULL
 - (b) p->next == NULL
 - (c) p == head
 - (d) p->next == head
12. Floyd's cycle detection algorithm works by using:
- (a) One pointer and a counter
 - (b) A stack of visited nodes
 - (c) Two pointers advancing at different rates
 - (d) Marking each visited node with a flag
13. Iterative reversal of a singly linked list typically requires how many auxiliary pointers?
- (a) One
 - (b) Two
 - (c) Three
 - (d) Four
14. The condition head != NULL && head->next == NULL indicates that the list has:
- (a) Zero nodes
 - (b) Exactly one node
 - (c) Exactly two nodes
 - (d) A cycle
15. Which C statement correctly allocates memory for one node?
- (a) p = malloc(struct node);
 - (b) p = (struct node*)malloc(sizeof(struct node));
 - (c) p = (struct node*)malloc(sizeof(p));
 - (d) p = new struct node;
16. The main purpose of using a dummy (header) node is to:
- (a) Store the length of the list
 - (b) Avoid special-case handling for insertion and deletion at the head
 - (c) Allow backward traversal
 - (d) Prevent memory leaks automatically
17. In a circular singly linked list, the next field of the last node points to:
- (a) NULL
 - (b) Itself

- (c) The first node
 - (d) The second-to-last node
18. Which of the following is a genuine disadvantage of a singly linked list compared to an array?
- (a) It cannot store duplicate values
 - (b) It does not support random (indexed) access
 - (c) Its size must be fixed at compile time
 - (d) It cannot store structures as data
19. If the head pointer of a singly linked list is accidentally overwritten without saving it, the result is:
- (a) A cycle is created in the list
 - (b) Loss of access to all nodes, causing a memory leak
 - (c) The list is automatically freed
 - (d) The last node becomes the new head
20. To find the middle node of a singly linked list in a single traversal, one may use:
- (a) One pointer and the node count
 - (b) A slow pointer and a fast pointer
 - (c) Recursion applied to the tail
 - (d) A backward pointer from the last node
21. After executing `p = head; while (p->next != NULL) p = p->next;` on a non-empty list, the pointer `p` refers to:
- (a) The head node
 - (b) NULL
 - (c) The last node
 - (d) The second-last node
22. Which of the following statements is **true** about a singly linked list?
- (a) Nodes must occupy contiguous memory locations
 - (b) Nodes may be scattered anywhere in memory
 - (c) The size must be declared in advance
 - (d) All nodes must store identical data values
23. Write the C structure declaration for a node of a singly linked list that stores an integer.
24. Consider the following code fragment. What is the output?
- ```
// List: 10 -> 20 -> 30 -> 40 -> NULL
struct node *p = head;
while (p != NULL && p->next != NULL) {
 printf("%d ", p->data + p->next->data);
 p = p->next;
}
```
25. The following function is intended to insert a node at the end of the list. Identify and correct the error(s).
- ```
void insertEnd(struct node *head, int x) {
    struct node *t = (struct node*)malloc(sizeof(struct node));
    t->data = x;
    t->next = NULL;
    struct node *p = head;
    while (p->next != NULL)
        p = p->next;
    p->next = t;
}
```

- 26.** Write an iterative function to reverse a singly linked list and return the new head pointer.
- 27.** Explain, with a diagram, the pointer adjustments required to delete a node that lies between two other nodes in a singly linked list.
- 28.** A singly linked list contains 5 -> 8 -> 12 -> 3 -> NULL. Show the state of the list after the following operations, performed in order: (i) insert 7 at the beginning, (ii) delete the node containing 12, (iii) insert 9 at the end.